

Predicting Customer Churn in the Telecommunication Industries Using Machine Learning Techniques

Adeline Makokha

Registration No: 191199

Strathmore Institute of Mathematical Sciences

Supervisor: Prof. Henry Muchiri

November, 2025



Table of Contents

- 1 Introduction
- 2 Literature Review
- 3 Methodology
- 4 Expected Outcomes

Introduction

- Customer churn is a critical challenge that threatens profitability, sustainability, and competitiveness.
- Globally, churn rates in telecommunications of above 30%, leading to substantial revenue losses (Khan2024).
- Retaining customers is five to seven times cheaper than acquiring new ones (Kotler et al., 2021).
- In Sub-Saharan Africa, telecommunications access drives digital inclusion and economic participation (GSMA, 2022).

Problem Statement

- The telecommunications sector faces rising competition and evolving customer expectations (Khan 2024), increasing the risk of churn.
- Existing churn prediction methods are largely reactive, (Jahromi 2020) relying on static indicators and manual analysis.
- These traditional approaches are time-consuming, inefficient, and prone to human bias.
- They fail to capture behavioral and socio-economic changes, (Khan 2024) leading to poor retention and revenue loss.

Research Objectives

Primary Objective

To identify key churn predictors in telecommunication industry.

Specific Objectives

- i. To describe the challenges and limitations of existing literature in churn prediction in the telecommunications industry.
- ii. To develop a machine learning based churn prediction model using telecommunications data.
- iii. To evaluate the performance of Machine Learning algorithms.
- iv. To deploy Machine Learning models in an interactive dashboard.

Literature Review : Telecommunication Context

- Rapid technological change, market liberalization, and shifting customer needs have heightened competition in telecoms (Chang2024). With converging services and lower switching costs, customer retention has become crucial. Firms must use data and digital tools to sustain loyalty.
- High churn rates (over 30%) threaten profitability and inflate acquisition costs (Khan2024). Machine learning models like Random Forest and Gradient Boosting achieved over 90% churn prediction accuracy. Key factors included poor network quality, pricing, and low engagement, highlighting the role of predictive analytics in driving retention.

Literature Review : Theoretical Framework- CRM

- Customer experience theory links digital interactions to satisfaction and loyalty (Smith2021). In telecoms, poor service experiences drive churn, while seamless digital engagement can cut it by up to 25%.
- Customer Relationship Management (CRM) frameworks harness customer data to personalize communication, offers, and rewards (Rahman2024). Rahman used interpretable machine learning on telecom data to link churn prediction with targeted promotions and loyalty programs. The study achieved high accuracy in identifying at-risk customers and showed that personalized engagement can raise retention by 20–30%, positioning CRM analytics as a proactive churn prevention tool.

Literature Review : Empirical evidence on churn prediction

- Nagarkar (2022) used XGBoost on behavioral and recharge data to analyze churn in Nepal's telecom sector, addressing the lack of regional data-driven insights. Recharge frequency and inactivity duration were key churn indicators, and model-guided actions reduced customer loss by 17%, showing the value of ensemble learning for actionable retention strategies.
- Poudel and Sharma (2024) enhanced churn prediction with interpretable, explainer-aware models achieving AUC above 0.90. Their use of feature attribution improved transparency in AI decisions, helping telecoms understand churn causes and design effective retention programs marking a shift toward explainable and trustworthy AI in customer analytics.

Literature Review : Relevance and Contribution

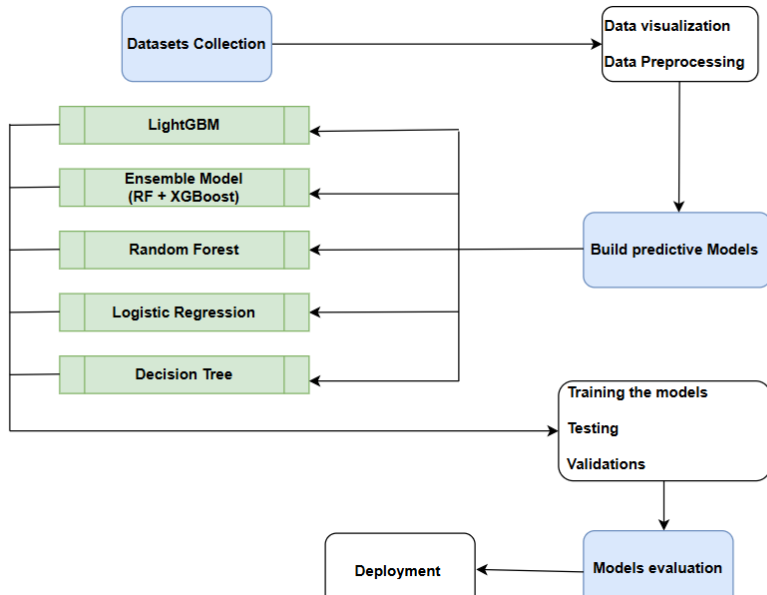
- Mwaura (2021) linked high telecom churn rates in Sub-Saharan Africa (above 30%) to reduced access to mobile finance, digital education, and SME growth. Using survey and secondary data, the study showed that frequent switching erodes trust and hinders digital inclusion, framing churn as both a business and development challenge.
- Shahabikargar and Kiani (2025) applied hybrid churn models integrating transactional, behavioral, and contextual data. Their approach improved accuracy by 10–15% over static models and enhanced transparency, showing how dynamic, explainable AI strengthens predictive power and managerial confidence in retention strategies.

- **Feature Limitations:** Available datasets mainly include structured CRM variables (billing, tenure, usage), but exclude important signals such as Total Revenue, Average Mobile Revenue, Average Fix Revenue, and ARPU. This narrows the feature space for building robust churn models.
- **Evaluation Metrics Bias:** Existing research often prioritizes accuracy and AUC, while overlooking profit-sensitive or ROI-based metrics like ARPU (Average Revenue Per User). This creates a gap between academic results and actionable business insights.
- **Interpretability Challenges:** Many ML models offer limited transparency, reducing business stakeholders understanding and trust in the results and translate them into real retention strategies.

Methodology: CRISP-DM Approach

- The study applies the CRISP-DM framework to guide a structured and scalable churn prediction process.
- Uses secondary anonymized public data from customer churn dataset containing real records from a leading Bulgarian telecommunication operator.
- Datasets cover demographics, contracts, billing, customer calling behavior and service usage for telecommunication customers.
- Enables cross-market comparison and validation of model performance.
- Data use aligns with global ethical and governance standards.

Methodology: Workflow



Methodology: Data Understanding and Cleaning

- A preliminary Exploratory Data Analysis (EDA) will summarize churn versus non-churn patterns using statistics, cross-tabs, and visualizations.
- Data cleaning will enhance quality by imputing missing values, correcting errors, and standardizing categorical/numeric variables.
- Outliers will be detected through statistical thresholds and visual diagnostics, then treated through capping, transformation, or removal.
- This preprocessing ensures a robust dataset for reliable predictive modeling.

- **Problem framing:** Customer churn is a binary classification task:

$$y_i = \begin{cases} 1, & \text{if CHURN} = 1 \text{ (customer churned)} \\ 0, & \text{if CHURN} = 0 \text{ (active customer)} \end{cases}$$

- **Comparative models:** Logistic Regression, Decision Tree, Random Forest, LightGBM, Ensemble models, and Logistic Regression with SMOTE to manage class imbalance.
- **Logistic Regression formulation:**

$$P(y = 1|x) = \frac{1}{1 + e^{-(\beta_0 + \sum_{j=1}^n \beta_j x_j)}}$$

where predictors include subscriber activity and revenue attributes.

- **Target interpretation:** $y_i = 1$ indicates customers lost within the observation period; $y_i = 0$ indicates retention.



Methodology: Machine Learning Modelling

- **Derived indicators:**

$$\text{ARPU} = \frac{\text{TotalRevenue}}{\text{Total_SUBs}}, \quad \text{ActiveRate} = \frac{\text{Active_subscribers}}{\text{Total_SUBs}}$$

$$\text{RevenueMix} = \frac{\text{AvgMobileRevenue}}{\text{AvgMobileRevenue} + \text{AvgFIXRevenue}}$$

- **SMOTE resampling (balancing minority churn class):**

$$x_{\text{new}} = x_i + \lambda(x_{nn} - x_i), \quad \lambda \in [0, 1]$$

- **Feature scaling:**

$$z = \frac{x - \mu}{\sigma}$$

applied to numeric fields (ARPU, ActiveRate, RevenueMix, etc.) to stabilize learning.

- **Encoding:** Categorical variables (CRM_PID_Value_Segment, EffectiveSegment, KA_name) encoded via One-Hot Encoding; redundant or highly correlated features removed to enhance model interpretability and efficiency.



Methodology: Evaluation and Deployment

- **Data split:** Models trained using an 80/20 stratified split to preserve the churn class balance:

$$\text{Train:Test} = 0.8 : 0.2, \quad P(y=1)_{\text{train}} \approx P(y=1)_{\text{test}}$$

- **Evaluation metrics:** Performance assessed using:

$$\text{Precision} = \frac{TP}{TP + FP}, \quad \text{Recall} = \frac{TP}{TP + FN}$$

$$\text{F1-Score} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}, \quad \text{ROC} = \int_0^1 \text{TPR}(\text{FPR}) d(\text{FPR})$$

Recall prioritized to minimize false negatives customers incorrectly predicted as non-churners.

- **Model deployment:** The best-performing model will be deployed through a RESTful API integrated with the CRM system, enabling churn probability outputs supporting targeted retention campaigns at scale.

$$\hat{P}(\text{CHURN} = 1|X_i)$$

Expected Outcomes

Technical Outcomes

- High model performance across key metrics (Recall, Precision, F1-score, ROC-AUC).
- Scalable architecture for seamless CRM integration and real-time predictions.

Operational Outcomes

- Proactive retention strategies to reduce revenue loss and improve loyalty.
- Optimized resource allocation targeting high-risk, high-value customers.

Broader Impact

- Provides a replicable framework for African telecommunication markets.
- Strengthens data-driven decision-making and digital inclusion.

References

- Chang, V., Hall, K., Xu, Q. A., Amao, F. O., Ganatra, M. A., & Benson, V. (2024). Prediction of customer churn behavior in the telecommunication industry using machine learning models. *Algorithms*, 17(6), 231. <https://doi.org/10.3390/a17060231>
- Communications Authority of Kenya. (2023). *Telecommunications market reports*. Retrieved June 9, 2025, from <https://ca.go.ke>
- GSMA Intelligence. (2022). *Mobile economy report: Sub-Saharan Africa 2022*. Retrieved June 9, 2025, from <https://www.gsma.com>
- Mwaura, E. T. (2021). *Effects of customer experience management on customer churn in the telco industry in Kenya* [Master's project, Kenyatta University]. Afribary. <https://afribary.com/works/effects-of-customer-experience-management-on-customer-churn-in-the-telco-industry-in-kenya>
- Nagarkar, P. (2022). A customer churn prediction model using XGBoost for the telecommunication industry in Nepal. *Procedia Computer Science*, 215, 652–661. <https://doi.org/10.1016/j.procs.2022.12.067>
- Rahman, M., Bista, R., & Poudel, K. (2024). Explaining customer churn prediction in the telecom industry using interpretable machine learning techniques. *Information Sciences Letters*, 36, 100043. <https://doi.org/10.1016/j.isl.2024.100043>
- Saha, L., Tripathy, H. K., Gaber, T., El-Gohary, H., & El-Kenawy, E. S. M. (2023). Deep churn prediction method for telecommunication industry. *Sustainability*, 15(5), 4543. <https://doi.org/10.3390/su15054543>
- Shahabikargar, M., Beheshti, A., Mansoor, W., Zhang, X., Foo, E. J., Jolfaei, A., Hanif, A., & Shabani, N. (2025). ChurnKB: A generative AI-enriched knowledge base for customer churn feature engineering. *Algorithms*, 18(4), 238. <https://doi.org/10.3390/a18040238>
- Tokmakov, D. (2024). *Customer churn dataset containing real records from a leading Bulgarian telecom operator, specifically for business customers (Version 1)* [Data set]. Mendeley Data. <https://doi.org/10.17632/nrb55gr66h.1>
- Usman-Hamza, F. E., Balogun, A. O., Capretz, L. F., Mojeed, H. A., Mahamad, S., Salihu, S. A., Akintola, A. G., Basri, S., Amosa, R. T., & Salahdeen, N. K. (2022). Intelligent decision-forest models for customer churn prediction. *Applied Sciences*, 12(16), 8270. <https://doi.org/10.3390/app12168270>
- Wagh, S., et al. (2024). Customer churn prediction in telecom sector using machine learning and deep learning techniques. *Array*, 21, 100310. <https://doi.org/10.1016/j.array.2023.100310>